

Needle Thoracostomy

Needle thoracostomy is a procedure in which an incision is made into the chest wall and pleural cavity to drain air or fluid. This procedure may be done either alone or in conjunction with needle decompression of the chest.

Indications for needle thoracostomy

- Tension pneumothorax not relieved with needle thoracostomy
- Suspected hemothorax causing hypotension or inability to ventilate
- Traumatic cardiac arrest

Contraindications

- No absolute contraindications for emergent thoracostomy but patient should be in extremis with shock, hypoxia or inability to ventilate – unrelated to airway management
- Use caution to prevent provider finger laceration, when patient has multiple rib fractures

Relative Contraindications

- Overlying skin infection
- Anticoagulation or bleeding disorder

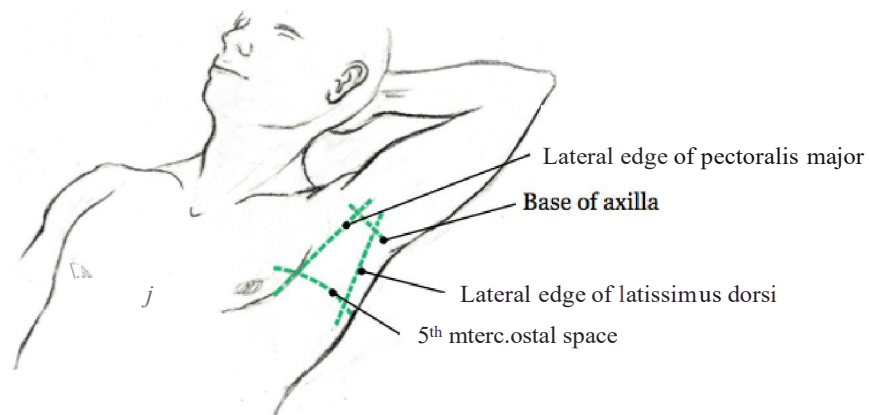
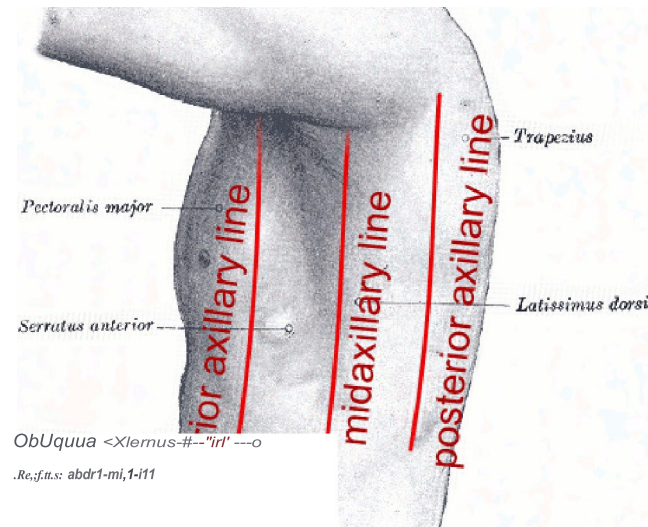
Equipment needed for finger thoracostomy

- Thoracostomy Needle

Anatomy for needle thoracostomy

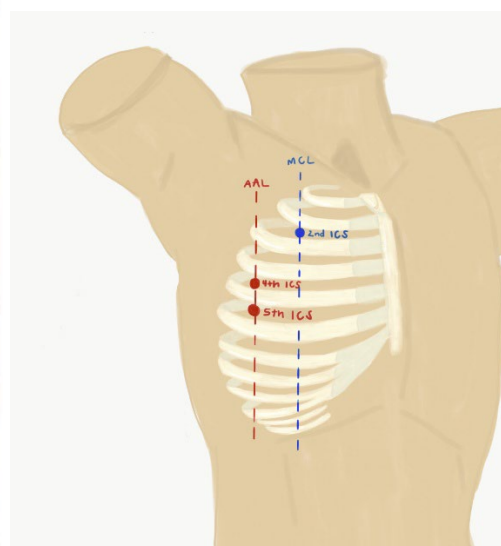
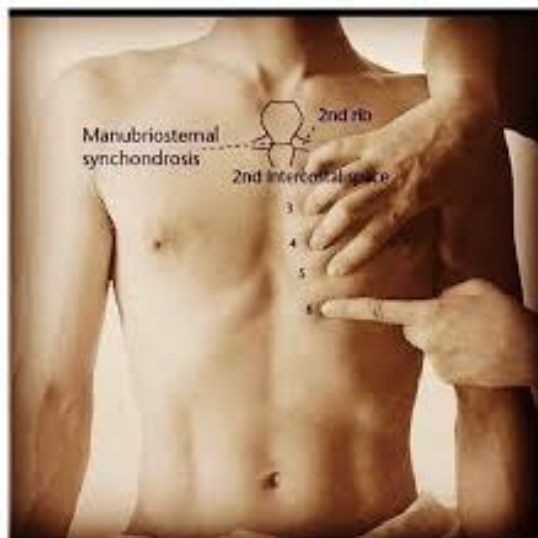
- Placement will be made at one of two locations:
 - The 5th intercostal space, between the anterior axillary and midaxillary line (see image)
 - The 2nd intercostal space, mid clavicular line (see image)

Fifth Intercostal Space



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2nd Intercostal Space



Needle Thoracostomy Procedure

- 1) Prepare equipment.
- 2) Place patient into supine position with ipsilateral arm abducted and externally rotated (so that hand is under or above the head).
- 3) Clean area with sterilizing prep, time permitting.
- 4) Identify the 5th intercostal space, between the anterior axillary and midaxillary line, or the 2nd intercostal space mid-clavicular line
- 5) Using a needle decompression spear scalpel tip, nick the skin to the inferior rib
- 6) Move your insertion hand index finger to 3cm away from the skin on the needle as marked
- 7) Hold your index finger in place as you insert the needle over the superior aspect of the rib into the indicated intercostal space until your insertion hand index finger makes contact with the skin stop insertion.
- 8) Remove the needle from the sheath
- 9) Angle the sheath superiorly as you slide the sheath into the pleural space
- 10) Leave the sheath in place